Rushil Shah

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Education

University of Southern California — Los Angeles, CA
B.S. in Computer Science and Business Administration — *Aug. 2021 – May 2025*

Experience

Bank of America — New York, NY
Software Engineer II, Digital Assets — *Aug. 2026 – Present*

- Architected a new digital asset platform from inception, shaping product and design direction while owning the Express backend, OpenShift CI/CD, and the marketing landing page used to present the platform across the bank
- Shipped 20 pages of the platform UI in 2 weeks with typed React, shadcn, RTK Query, and Mock Service Worker (MSW), serving operations, production support, management, and business users across Risk and Collateral
- Led enterprise SSO integration using OAuth 2.0 Authorization Code + PKCE, adding role-based MFA and enforcing entitlement-driven authorization through an Express server managing sessions and tokens
- Built a GitHub Copilot agent framework from scratch (Jira, backend, frontend, testing agents) synced via a Bash script, adopted across the entire Digital Assets division and cutting ticket-to-PR time to same-day
- Established the team's Playwright testing framework from scratch, building 60 smoke, mocked, and end-to-end tests with a coverage matrix for client onboarding, integrated into Jenkins CI with a 5-minute full-suite run

Software Engineer I — *Jul. 2025 – Aug. 2026*

- Consolidated 5+ tools into one Python/Dash dashboard for market risk EOD batch monitoring, pulling from Sybase, APIs, web scraping, and email and automating alerts and resubmissions, saving 20+ hours weekly
- Automated the weekly market risk stability deck as a scheduled Python batch job, visualizing report delays and root causes for the Risk Stability team and eliminating 10+ hours of weekly manual work

Trojan Marketing Group — Los Angeles, CA
Director of Technology — *Feb. 2024 – May 2025*

- Launched a React app for a beverage client's live event using Spotify OAuth and Grok, recommending drinks from the client's menu based on each attendee's top artists and tracks for 300+ attendees
- Directed a 12-member tech team across 4 client accounts, running sprints, mentoring members, presenting technical direction to clients, and tying each build to the client's marketing campaign

NASA Jet Propulsion Laboratory — Pasadena, CA
Software Development Intern — *Sep. 2023 – May 2024*

- Built a migration QA tool in Python and Playwright that compared old WordPress pages against their science.nasa.gov versions, automatically flagging missing content, images, broken links, and structural differences
- Developed a React/TypeScript dashboard on FastAPI with batch CSV validation of URL pairs, tracking pass rates, common issue types, and validation history for the migration team
- Remediated WCAG 2.1 accessibility and layout defects during NASA's migration of 200+ WordPress pages to science.nasa.gov, fixing CMS template-level issues and flagging 50+ defects for the content team

Projects

On-Chain Commish | *[GitHub]* — Oct. 2026 – Present

- Developing a fantasy league treasury on Ethereum that escrows buy-ins and pays winners from ESPN standings posted on-chain by a daily bot, using Solidity, Foundry, React, wagmi, viem, RainbowKit, and Alchemy
- Implementing majority-vote disputes, failure-path refunds, and pull-pattern payouts so no one controls or locks funds; deployed to Sepolia and verified on Etherscan, with 142 unit, fuzz, and reentrancy tests (∼98.7% coverage)

Python Redis from Scratch | *[GitHub]* — Oct. 2025 – Nov. 2025

- Engineered a master-replica synchronization protocol featuring a multi-step handshake (PING, REPLCONF, PSYNC) and real-time command propagation to keep replicas consistent
- Built a multi-threaded TCP server in Python using threading.Lock and Condition variables to manage shared state and enable blocking operations like BLPOP and XREAD BLOCK

Skills

Languages: Python, TypeScript, JavaScript, Java, SQL, C++, Solidity
Frameworks: React, Next.js, Redux, Express, FastAPI, Flask, Dash, Tailwind CSS, shadcn, wagmi, viem
Databases & AI: PostgreSQL, Redis, Sybase, ChromaDB, XGBoost, LangChain, LLM APIs
Tools: Docker, Jenkins, OpenShift, AWS, Git, Foundry, GitHub Actions, Playwright, k6, Pytest